



UNIVERSIDAD AUTONOMA DE NUEVO LEON



**Facultad de Ciencias Físico Matemáticas
(FCFM)**

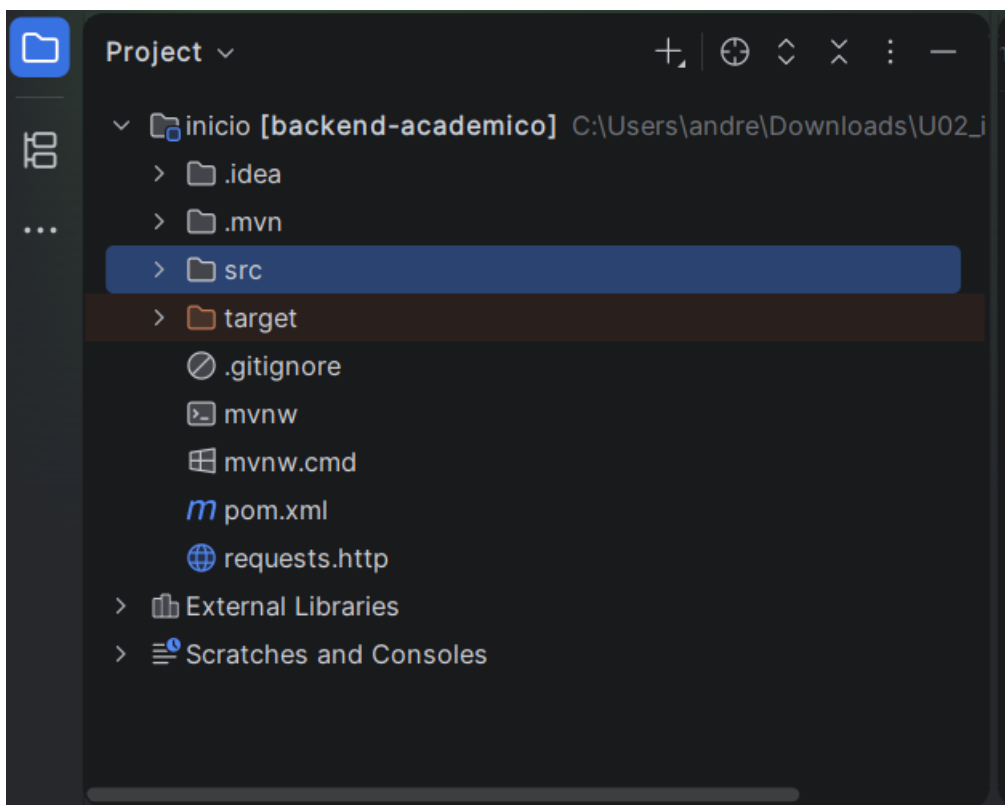
**Unidad de Aprendizaje:
Desarrollo web: Tecnologías Servidor**

Unidad 02

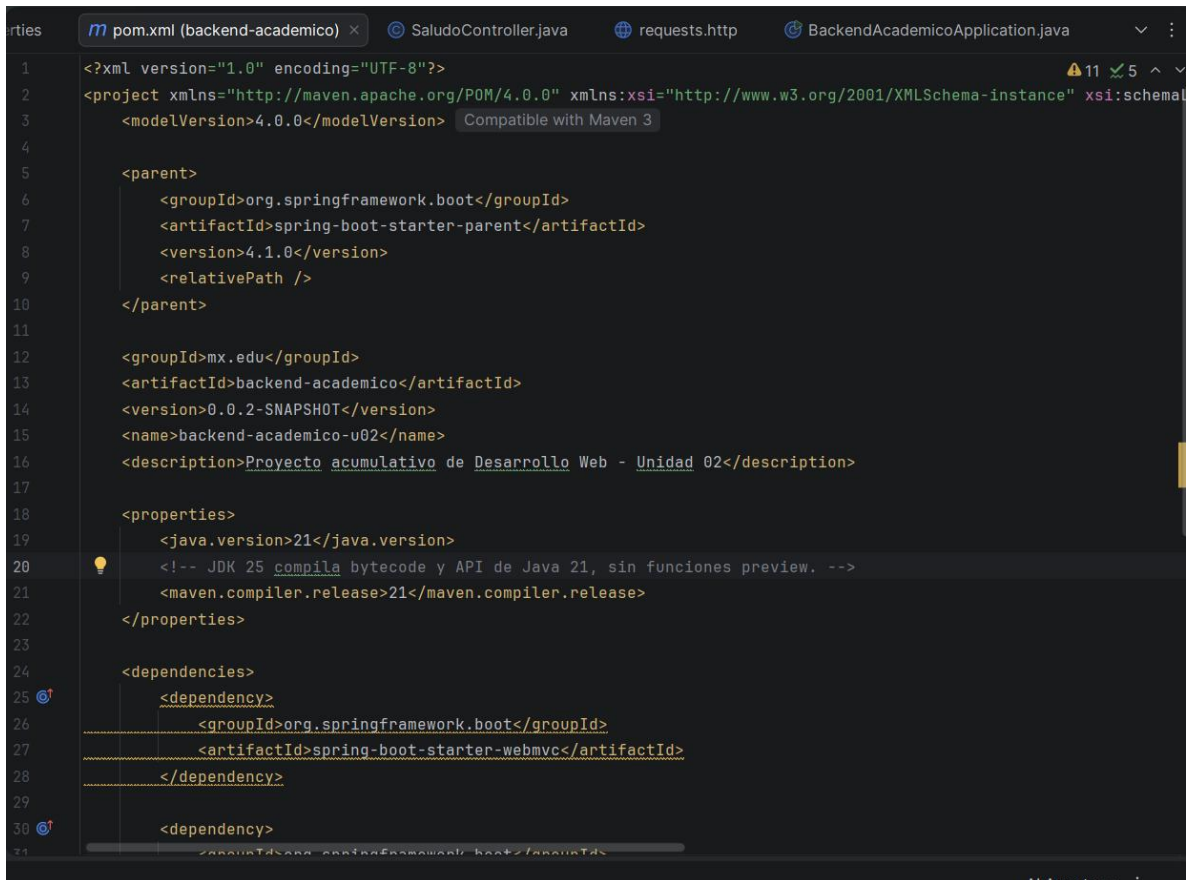
Nombre:	Andrés Castro Hernandez
Matricula:	2069855
Grupo:	031
Profesor:	Gustavo Adolfo Rodriguez Martinez
Carrera:	LCC

Monterrey N.L. a; octubre de 2026

Proyecto:

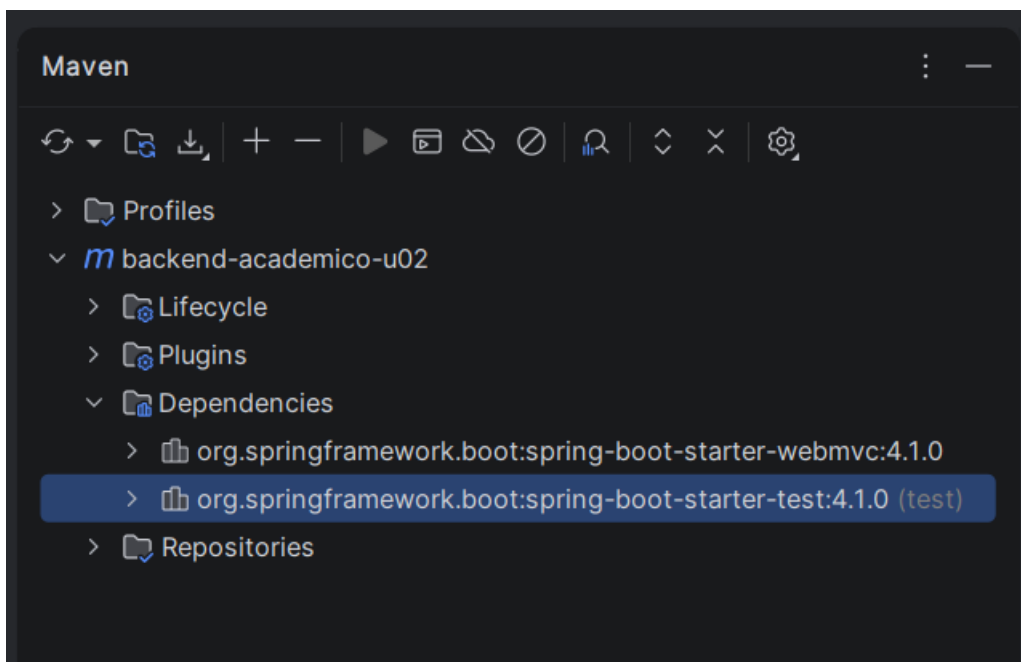


Paso 1:

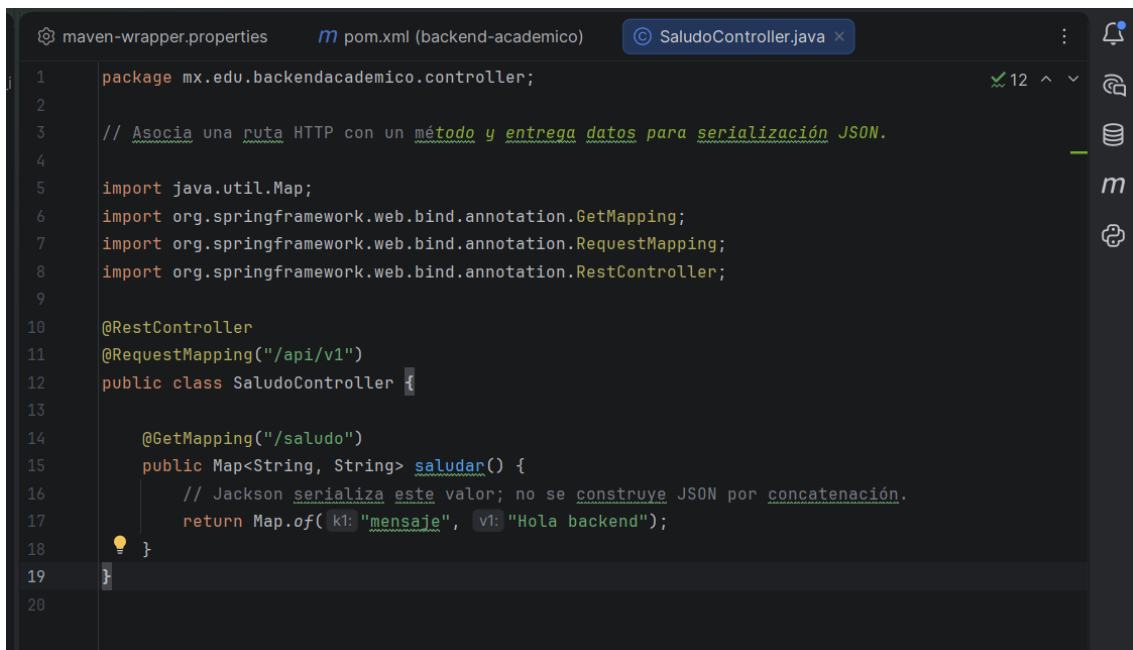
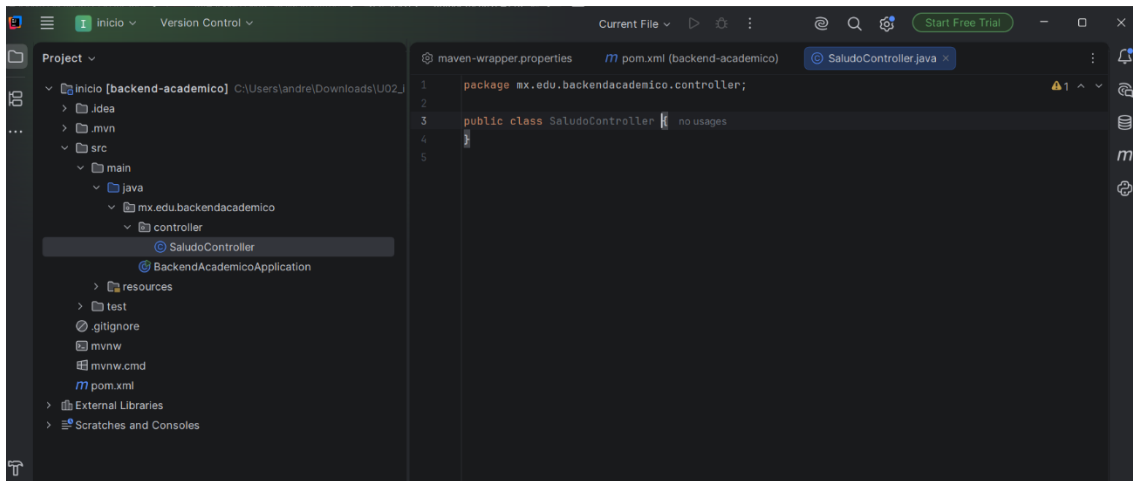


The screenshot shows an IDE with the 'pom.xml' file for 'backend-academico' open. The file contains the following XML structure:

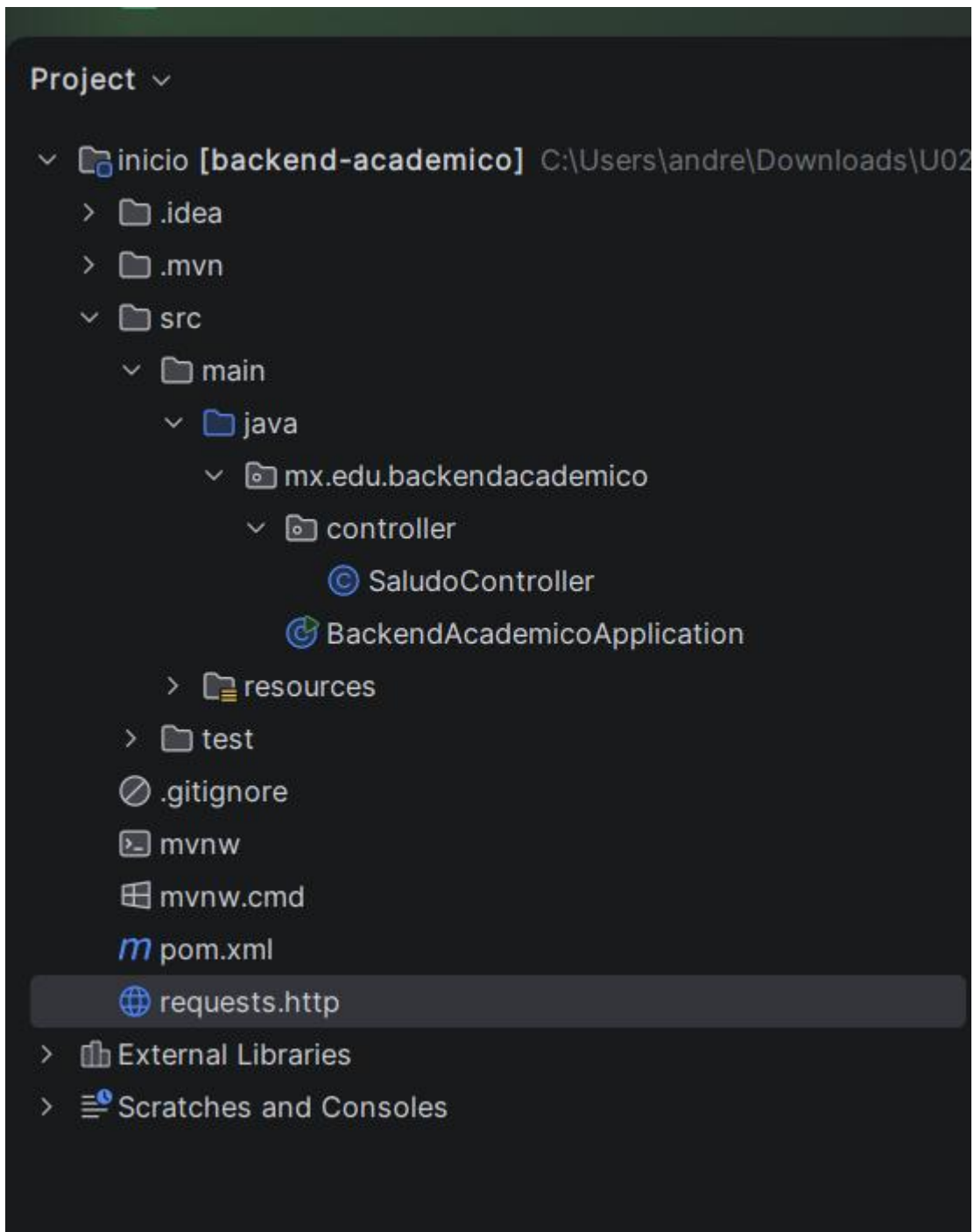
```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>4.1.0</version>
    <relativePath />
  </parent>
  <groupId>mx.edu</groupId>
  <artifactId>backend-academico</artifactId>
  <version>0.0.2-SNAPSHOT</version>
  <name>backend-academico-u02</name>
  <description>Proyecto acumulativo de Desarrollo Web - Unidad 02</description>
  <properties>
    <java.version>21</java.version>
    <!-- JDK 25 compila bytecode y API de Java 21, sin funciones preview. -->
    <maven.compiler.release>21</maven.compiler.release>
  </properties>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-webmvc</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
    </dependency>
  </dependencies>
</project>
```

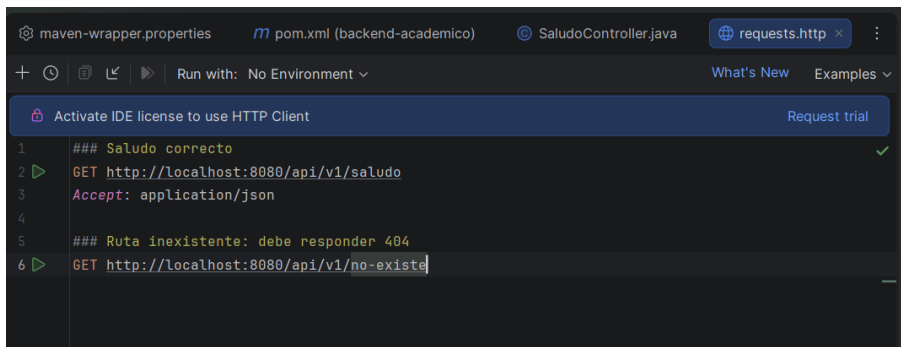


Paso 2:



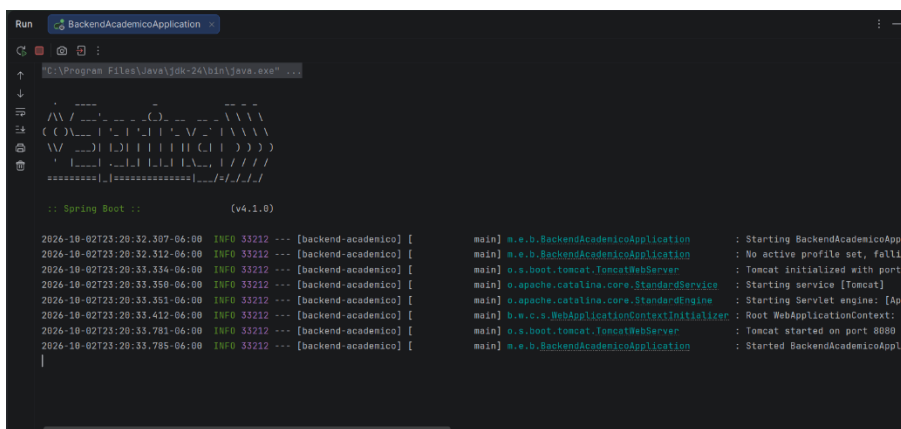
Paso 3:



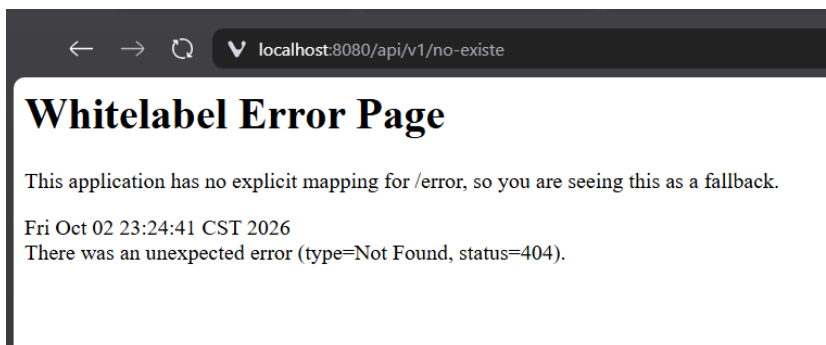
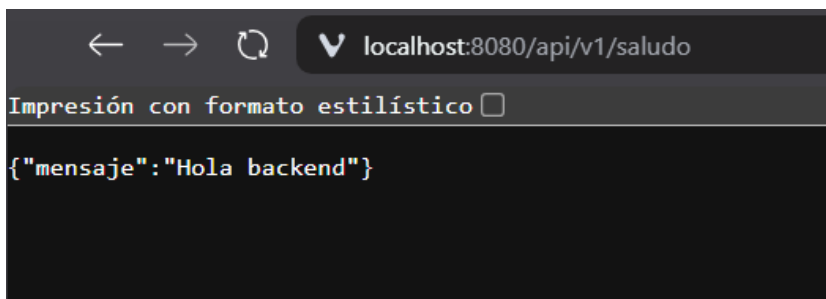


```
1  ### Saludo correcto
2  GET http://localhost:8080/api/v1/saludo
3  Accept: application/json
4
5  ### Ruta inexistente: debe responder 404
6  GET http://localhost:8080/api/v1/no-existe
```

Cierre de la construcción guiada:



```
Run BackendAcademicoApplication
C:\Program Files\Java\jdk-24\bin\java.exe ...
... Spring Boot ... (v4.1.0)
2026-10-02T23:20:32.307-06:00 INFO 33212 --- [backend-academico] [main] m.e.b.BackendAcademicoApplication : Starting BackendAcademicoAppli
2026-10-02T23:20:32.312-06:00 INFO 33212 --- [backend-academico] [main] m.e.b.BackendAcademicoApplication : No active profile set, falling
2026-10-02T23:20:33.334-06:00 INFO 33212 --- [backend-academico] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8
2026-10-02T23:20:33.350-06:00 INFO 33212 --- [backend-academico] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-10-02T23:20:33.351-06:00 INFO 33212 --- [backend-academico] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apac
2026-10-02T23:20:33.412-06:00 INFO 33212 --- [backend-academico] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: in
2026-10-02T23:20:33.791-06:00 INFO 33212 --- [backend-academico] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (h
2026-10-02T23:20:33.795-06:00 INFO 33212 --- [backend-academico] [main] m.e.b.BackendAcademicoApplication : Started BackendAcademicoAppl
```



Se comprobó que los archivos y los resultados fueran iguales en el archivo de solución 👍

```
.\mvnw.cmd test
```

Inicio con:

```
.\mvnw.cmd spring-boot:run
```

GET

Reto:

Modificación de SaludoController

```
BackendAcademicoApplication.java SaludoController.java x
1 package mx.edu.backendacademico.controller;
2
3 // Asocia una ruta HTTP con un método y entrega datos para serialización JSON.
4
5 > import ...
10
11 @RestController
12 @RequestMapping("/api/v1")
13 public class SaludoController {
14
15     @GetMapping("/saludo")
16     public Map<String, String> saludar(
17         @RequestParam(defaultValue = "Mundo") String nombre) {
18         // Jackson serializa este valor; no se construye JSON por concatenación.
19         return Map.of( k1: "mensaje", v1: "Hola " + nombre);
20     }
21 }
```

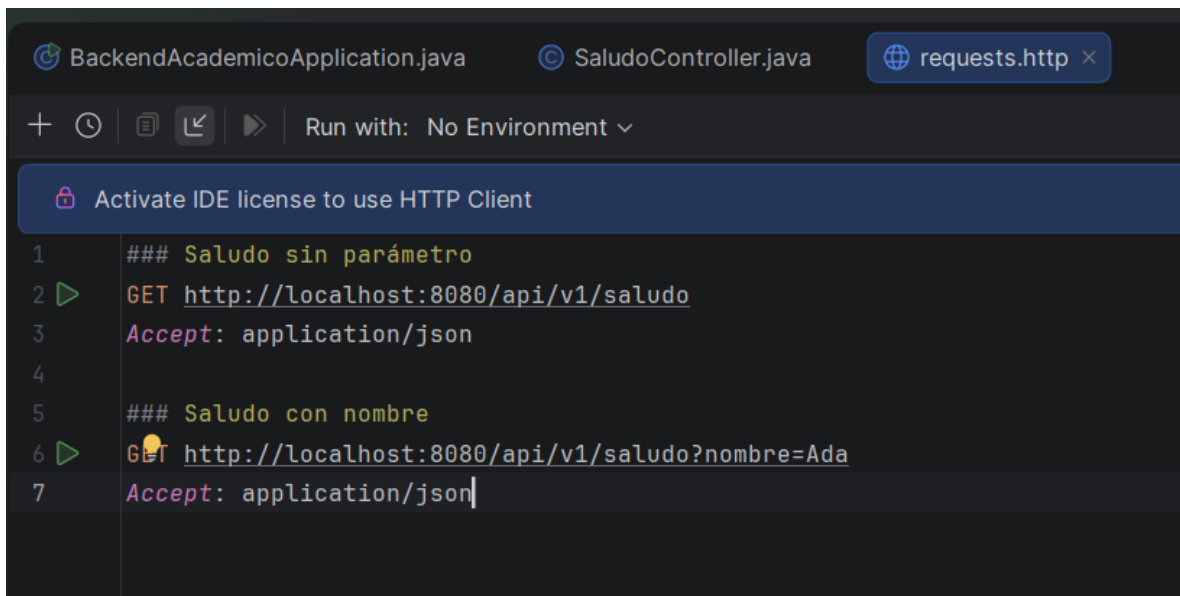
Prueba con `.\mvnw.cmd test`

```
Terminal Local x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

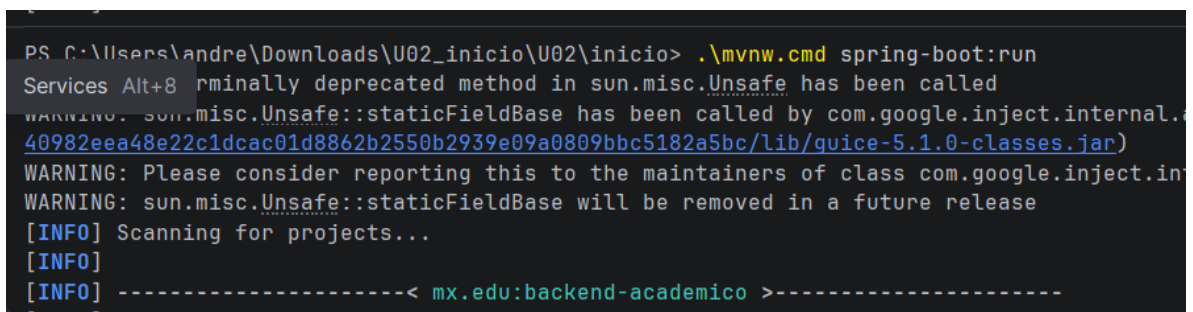
PS C:\Users\andre\Downloads\U02_inicio\U02\inicio> .\mvnw.cmd test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClass
40982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/quice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.Hid
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
```

```
Terminal Local x + v
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.578 s
[INFO] Finished at: 2026-10-03T00:27:21-06:00
[INFO] -----
PS C:\Users\andre\Downloads\U02_inicio\U02\inicio>
```

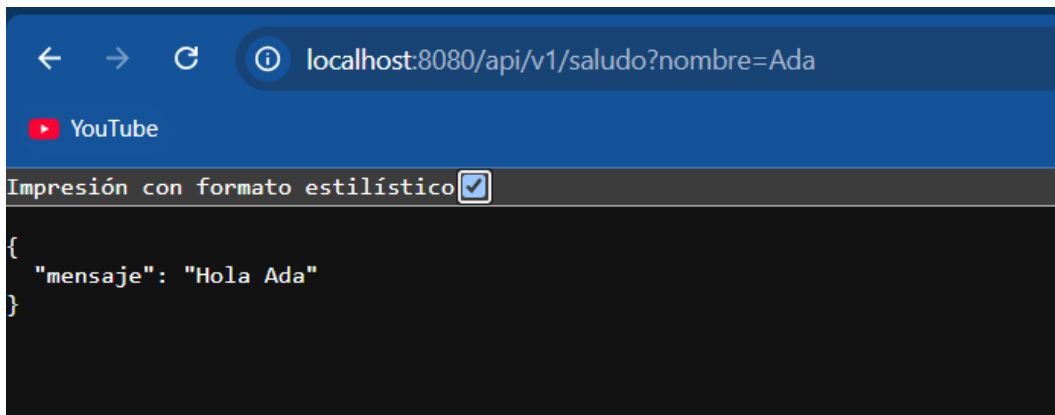
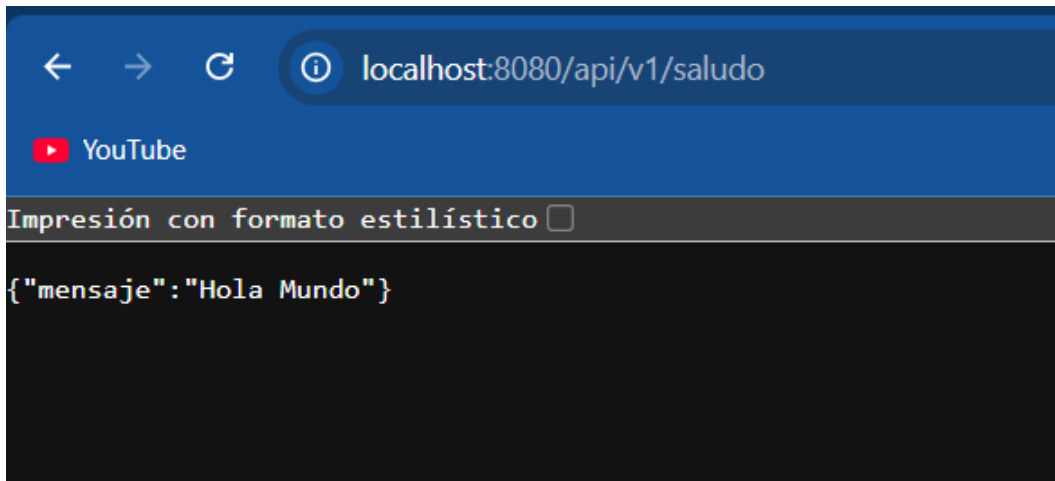
Modificacion de requests.http



Inicio con .\mvnw.cmd spring-boot:run



Pruebas:



El parámetro nombre se resuelve durante el procesamiento de la petición por Spring MVC. La anotación `@RequestParam` indica a Spring que debe obtener el parámetro nombre de la URL. Si el parámetro no está presente, se utiliza el valor definido en `defaultValue`, que en este caso es "Mundo".